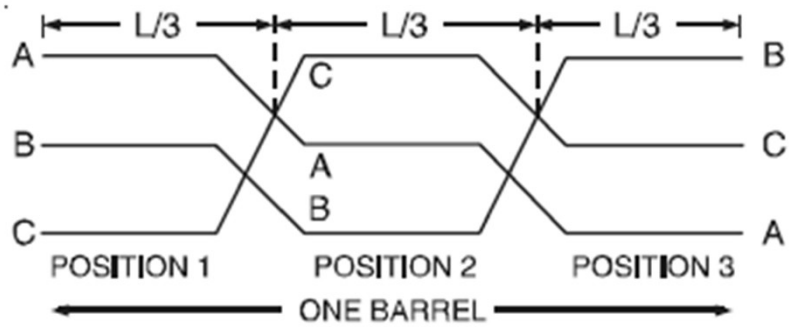
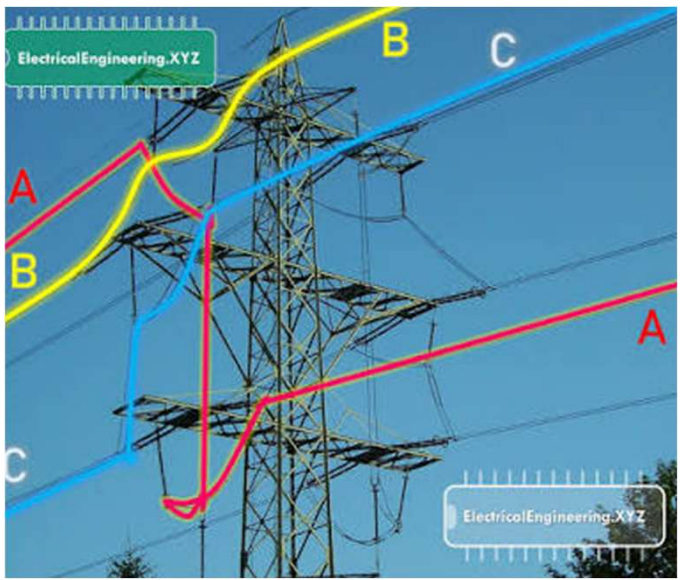
Transposition of a 3-Phase Line





$$\mathbf{L} = \frac{L_A + L_B + L_C}{3}$$

$$= 2 \times 10^{-7}\left[\log_e \frac{1}{r'} + \frac{1}{3}\left(\log_e \sqrt{d_1 d_3} + \log_e \sqrt{d_1 d_2} + \log_e \sqrt{d_2 d_3}\right)\right.$$

$$\left. + \frac{j\sqrt{3}}{3}\left(\log_e \sqrt{\frac{d_1}{d_3}} + \log_e \sqrt{\frac{d_2}{d_1}} + \log_e \sqrt{\frac{d_3}{d_2}}\right)\right] \text{ H/m}$$

$$= 2 \times 10^{-7}\left[\log_e \frac{1}{r'} + \log_e \sqrt[3]{d_1\, d_2\, d_3} + 0\right]$$

$$= 2 \times 10^{-7} \log_e \frac{\sqrt[3]{d_1\, d_2\, d_3}}{r'} \text{ H/m} \qquad \ldots(5.21)$$